MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL,
FLIP-FLOPS, CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/325C, dated 24 September 1987 and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, I_{IL}, data, device type 03, maximum limits column: Delete "-360" and substitute "-400".

TABLE I, I_{IL}, enable, device type 04, maximum limits column: Delete "-400" and substitute "-360".

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FIGURE 4, note 4: Delete and substitute "4. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V."

FIGURE 4, note 5: Delete and substitute "5. C_L = 50 pF ±10 percent (including jig and probe capacitance)."

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TABLE III, footnote 1/, O/C parameter, circuit F: Delete "-160/-400" and substitute "-400 max".

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TABLE III, I_{CC}, maximum limits column: Delete "40" and substitute "42".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	Reprinted without change
12	12
21	21
22	22
41	41
42	Reprinted without change

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TABLE III, footnote 2/, O/C parameter, circuit F: Delete "-160/-400" and substitute "-400 max".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1281)

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Device type 01

Inputs			Output
Clear	Clock	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated input conditions were established

Device type 02

Output control	Enable G	D	Output
L	H	H	H
L	H	L	L
L	L	X	Q_0
H	X	X	Z

H = high level (steady state)
L = low level (steady state)
X = irrelevant
Q_0 = the level of Q before the indicated steady state input conditions were established
Z = high impedance state

FIGURE 2. Truth tables.

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Device type 03

Output control	Clock	D	Output
L	↑	H	H
L	↑	L	L
L	L	X	Q_0
H	X	X	Z

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established
Z = high impedance state

Device type 04

Inputs			Output
$\overline{G}$	Clock	Data	Q
H	X	X	Q_0
L	↑	H	H
L	↑	L	L
X	L	X	Q_0

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established

FIGURE 2. Truth tables - Continued.

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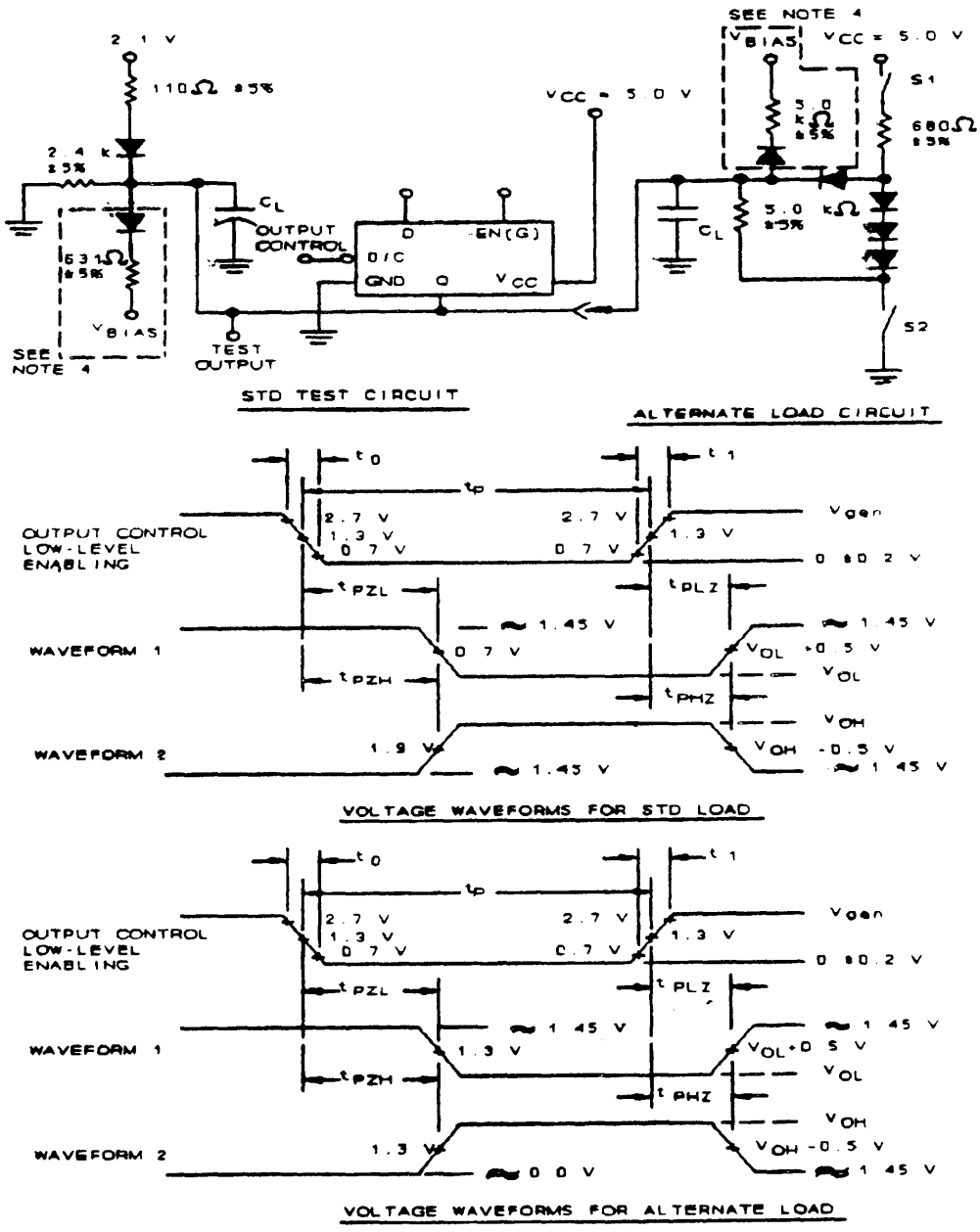


FIGURE 4. Three-state switching test circuit and waveforms for device type 02 - Continued.

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NOTES:
1. All diodes are 1N3064, or equivalent.
2. Output control input pulse characteristics: V_{gen} = 3 ±0.2 V, t_0 ≤ 6 ns, t_1 ≤ 15 ns, t_p (input) ≥ 200 ns, and PRR ≤ 1 MHz.
3. For the standard test circuit, C_L = 50 pF ±10 percent. For the alternate load circuit, C_L = 50 pF ±10 percent for t_{PZL} and t_{PZH} and C_L = 15 pF minimum for t_{PLZ} and t_{PHZ}. All capacitive load values include jig and probe capacitance.
4. The diodes and resistors shown within the dotted areas are optional. When the diodes and resistors are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} test, V_{BIAS} shall be -0.6 V.
5. When using tne alternate load circuit, the appropriate switch positions for the three-state tests are listed in the table below.

Test	S1	S2
t_{PZL}	Closed	Open
t_{PLZ}	Closed	Closed
t_{PZH}	Open	Closed
t_{PHZ}	Closed	Closed

FIGURE 4. Three-state switching test circuit and waveforms for device type 02 - Continued.

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4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C, minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II, herein.

4.4.3 Group C inspections. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition D, E, or F, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for JAN marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - - - Ground zero voltage potential

V_{IN} - - - - - - - - - - - - - - - - - Voltage level at an input terminal

f_{MAX} - - - - - - - - - - - - - - - - - Maximum clock frequency

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuits types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS273
02	54LS373
03	54LS374
04	54LS377

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